

153.8 Tb/s O-Band Coherent Transmission over SMF with Low-Complexity DSP

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Abstract: We demonstrate record 153.8 Tb/s (50.3 km) and 141.1 Tb/s (75.5 km) O-band (16.5 THz) WDM transmission with simplified DSP using geometrically-shaped 256QAM without chromatic dispersion compensation and a short 7-tap adaptive equaliser.

1. Introduction

Optical fibre networks form the backbone of modern digital infrastructure, enabling global connectivity and supporting the exponential growth of data traffic driven by cloud computing, artificial intelligence, and data-centre interconnects [1, 2]. To meet this growing demand, spectral regions beyond the conventional C-band are being explored to extend the capacity of single-mode fibres further [3]. Among these, the O-band, centred at 1310 nm, has emerged as a promising candidate due to its wide optical bandwidth of approximately 17.5 THz and the intrinsically low chromatic dispersion of standard single-mode fibre (SSMF) across this range. The near-zero chromatic dispersion reduces the need for complex electronic dispersion compensation (EDC) and simplifies transceiver designs. In addition, the O-band is spectrally separated from the C-band, which effectively suppresses inter-channel stimulated Raman scattering (ISRS) between them in multi-band transmission systems. Although very promising, the O-band transmission is limited by higher fibre loss, approximately 0.3 dB/km due to increased Rayleigh scattering, compared to 0.15–0.2 dB/km in the C-band, as well as increased nonlinear distortions from the combination of a higher nonlinear coefficient and the low dispersion, where near phase matching condition enhances four-wave mixing (FWM) efficiency and significantly increases the multi-channel interference (MCI) [4, 5].

The recent development of bismuth-doped fibre amplifiers (BDFAs) has enhanced the feasibility of O-band communications, as BDFAs can efficiently amplify the entire O-band spectrum with a single amplifier, eliminating the need for parallel amplifiers and wavelength-division multiplexing (WDM) couplers [6]. In comparison, covering a similar bandwidth across the S, C, and L-bands typically requires three amplifier types, while the E-band is also wide but limited by fibre water peak absorption at short wavelengths. Previous work has reported transmission of 106 Tb/s over an 80.4 km fibre link, using BDFAs and a 16.4 THz spectral range [7], demonstrating the potential of high-capacity O-band coherent systems.

In this work, we investigated the impact of digital signal processing to transmit O-band (16.5 THz) signals over 50.3 km and 75.5 km of Corning® SMF-28® ULL fibre, achieving highest throughput values to date, namely 153.8 Tb/s and 141.1 (33% increase on the previous record), with only a 7-tap equaliser. Increasing the equaliser length to 31 taps raised the throughput to 156.1 Tb/s and 143.4 Tb/s, whilst the inclusion of electronic dispersion compensation (EDC) yielded no additional benefit for 32 GBaud signals over 75.5 km. These results demonstrate the potential of the O-band transmission, supported by BDFAs and simplified DSP, for high-throughput, low-complexity coherent systems.

2. Experimental Setup

Figure 1(a) shows the O-band experimental transmission setup with 501×32 GBaud WDM channels. Three independent tunable laser sources (with linewidths below 10 kHz), generated the 33 GHz-spaced carriers. The central laser was used for the channel under test (CUT) and the other two were combined and used as the neighbouring channels. The channels were then split by a polarisation-maintaining 3 dB coupler, and each arm was independently modulated by a 23 GHz bandwidth IQ-MZM to generate dual-polarisation 32 GBaud signals with 1%-roll-off root-raised cosine filters. Geometrically-shaped (GS) 256QAM [8] signals were generated, and linear digital pre-distortion [9] was applied. The modulators were driven by 92 GS/s arbitrary waveform generators (AWGs) with an ENOB of 5 bits. After modulation, the signals were combined with a polarisation beam combiner (PBC) to form dual-polarisation QAM signals. A variable optical attenuator (VOA) was used to set the CUT power equal to that of each neighbouring channel. After combining the CUT and neighbouring channels, BDFAs were used to boost the signal. Spectrally-shaped amplified spontaneous emission (SS-ASE) was used to emulate the remaining channels over the full O-band (1265–1355 nm). The ASE source consisted of a superluminescent diode (SLD) and a BDFA, spectrally shaped by a wavelength-selective switch (WSS), also used to carve a notch at the wavelengths of the three channels [10].

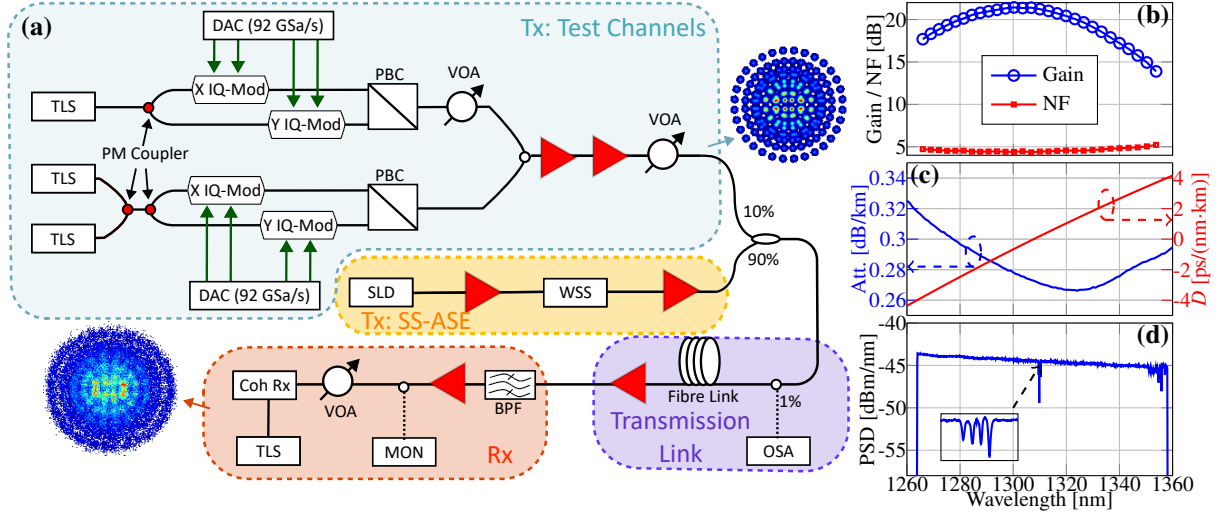


Fig. 1: (a) Experimental setup. (b) BDFAs gain and noise figure. (c) Fibre attenuation and dispersion parameters. (d) Transmission spectrum with zoomed inset.

The BDFAs consisted of bismuth-doped fibres (BDF) with lengths ranging from 120 m to 220 m, backwards-pumped at 1150 nm with optical power between 2 W and 3 W. All BDFAs used in the experimental setup had similar specifications, providing over 20 dB of gain and noise figures (NFs) between 4 dB and 5.5 dB across the 1265–1355 nm range for different input powers. Their design follows that described in [6]. An SS-ASE source and a WSS emulated 30×100 GHz WDM channels to characterise the BDFAs. Figure 1(b) reports the gain and NF versus wavelength at -3 dBm input for the channel-booster BDFAs.

A 1% tap and an optical spectrum analyser (OSA) were used to monitor and enable balancing of the powers of the three channels and the SS-ASE noise. The launch power (LP) was 21.7 dBm (approximately -5 dBm/ch) and two different link lengths of 50.3 km or 75.5 km of Corning® SMF-28® ULL fibre were used, with attenuation and dispersion depicted in Fig. 1(c). We note that this fibre, while compliant with the G.654.C specification allowing cable cutoff up to 1530 nm, has previously been demonstrated to successfully support O-band transmission with negligible penalty [11, 12] and our results here further confirm that capability.

A tilt of 2 dB across the signal band was applied to the transmitted spectrum to compensate for the higher losses in the shorter wavelengths and the inter-channel stimulated Raman scattering (ISRS) effect, as shown in Fig. 1(d). After transmission, the fibre attenuation was compensated for by an in-line BDFAs and the CUT was selected by an optical bandpass filter (BPF). A combination of a pre-amplifier, monitor (MON), and VOA was employed to maintain a constant received power of 1 dBm at the photodetectors, to ensure operation within their optimal sensitivity range.

The coherent receiver consisted of a 90° hybrid, four 70 GHz balanced photodetectors, and a 33 GHz digital sampling oscilloscope with 8 bit ADCs. The oscilloscope bandwidth on this occasion was lower than that of the photodetectors due to practical constraints, as a single instrument was used to capture all four tributaries simultaneously, without employing RealEdge technology. A Keysight O-band ECL with a linewidth below 10 kHz served as the local oscillator (LO). Following coherent detection, pilot-assisted DSP was applied using a pilot sequence of length of 2^{10} and an insertion rate of 1/32 [13]. We swept the CUT and its neighbouring channels across the band to measure throughput at every CUT position in the WDM grid. The transceiver back-to-back SNR was approximately 17.5 dB for all channels across the entire spectrum. The throughput was calculated from the generalised mutual information (GMI) and after decoding using DVB-S2X low-density parity check codes [14].

3. Experimental Results and Discussion

The received WDM signals over 50.3 km and 75.5 km were processed after coherent detection with equalisers of 3 to 71 taps, with and without EDC. Figures 2(a) and Fig. 2(b) show the total throughput, calculated from the GMI, for different number of taps. For 50.3 km, a 3-tap equaliser limits the throughput to 103.4 Tb/s (112.2 Tb/s with EDC). Increasing the number of taps from 5 to 31 raises throughput from 151.3 Tb/s to 156.1 Tb/s, while applying EDC brings no further gain. For longer filters, performance degrades as no residual impairment remains to be compensated, and the MIMO structure accumulates noise. Similar behaviour occurs at 75.5 km: a 5-tap equaliser yields 138.5 Tb/s without EDC, rising to 143.4 Tb/s with 31 taps; EDC has a negligible effect for ≥ 5 taps.

Figures 2(c) and 2(d) depict the throughput per WDM channel for 5-, 7-, and 31-tap equalisers at 50.3 km and 75.5 km. For both distances, shorter wavelengths degrade due to higher fibre attenuation (Fig. 1(b)), and both spectrum edges suffer from component limits (lower laser power, higher PM coupler/PBC loss and crosstalk,

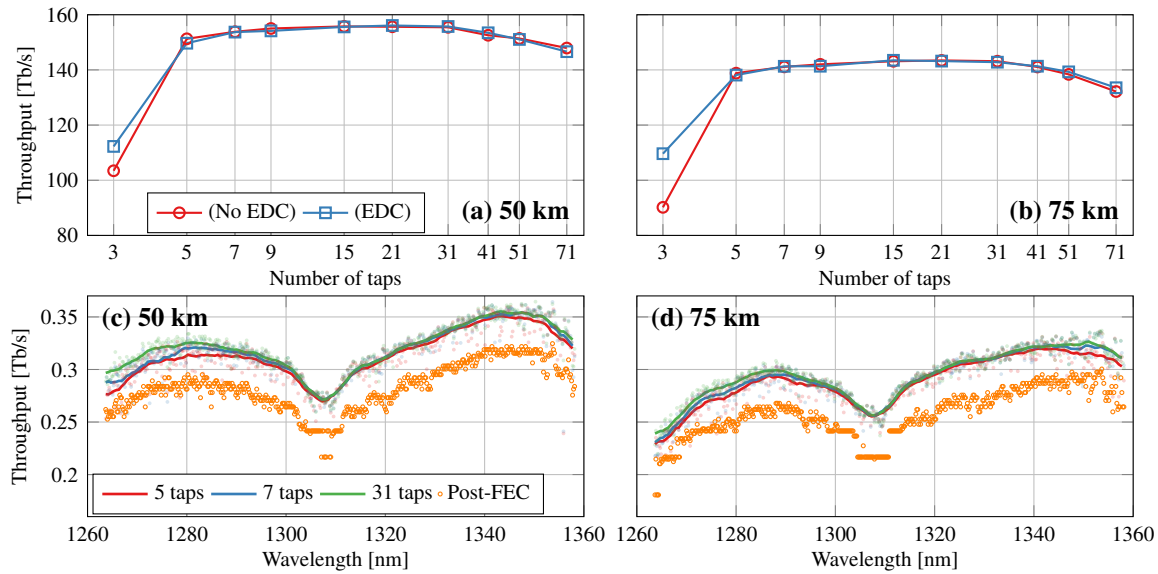


Fig. 2: Throughput derived from GMI for various equaliser tap numbers, in log scale, with and without EDC, for (a) 50 km and (b) 75 km. Throughput derived from GMI for each WDM channel with 5, 7, and 31 taps, and post-FEC data rate for 7 taps, for (c) 50 km and (d) 75 km. reduced amplifier gain). Longer wavelength channels perform best, due to the lower attenuation and a reduction in nonlinearity. A dip near 1307 nm (zero dispersion) indicates dominant MCI, with a 30% lower throughput compared to the best-performing channels.

For both transmission distance cases, shown in Figs. 2(c) and 2(d), a comparison between three equaliser configurations (5, 7, and 31 taps) reveals a similar overall behaviour. Around the zero-dispersion region (1307 nm), all configurations exhibit comparable performance. However, for wavelengths farther from 1307 nm, the magnitude of the dispersion can reach 4 ps/(nm·km), as shown in Fig. 1(b). In such cases, using a longer equaliser provides a noticeable performance improvement. The per-channel throughput gain is approximately 20 Gb/s per channel at wavelengths far from the zero-dispersion wavelength when increasing MIMO tap lengths from 5 to 31. Considering the trade-off between performance and complexity, the 7-tap equaliser is assumed in the following discussion.

Figures 2(c) and 2(d) also show the post-FEC rate after decoding for a 7-tap equaliser. For the 50.3 km link, over which the GMI data rate was 153.8 Tb/s, a post-FEC rate of 140.7 Tb/s was achieved. For 75.5 km, a post-FEC rate of 128.4 Tb/s was obtained, while the corresponding GMI data rate was 141.1 Tb/s. In both cases, a larger gap between the post-FEC rate and GMI is observed in the lower-SNR regions, where the post-FEC data rate tends to flatten. This behaviour results from employing a fixed GS-256QAM modulation format across all WDM channels. With an adaptive configuration (using lower-order modulation formats in low-SNR regions), the coding efficiency could be further improved. Overall, the post-FEC rate was approximately 91% of the GMI.

4. Conclusions

A 16.5 THz O-band signal was transmitted over two distances, 50.3 km and 75.5 km, of Corning® SMF-28® ULL fibre links, achieving throughputs of 153.8 Tb/s and 141.1 Tb/s, respectively, using a 7-tap equaliser. The total throughput could be increased by 2.3 Tb/s (1%) with longer equaliser length of 31 taps, a configuration commonly used in the C band and other bands. No significant improvement using additional EDC was observed for 32 GBaud signals over the 75.5 km link. To the best of our knowledge, this represents the highest throughput O-band transmission reported to date. The wide O-band spectral region, supported by a single type of amplifier (BDFA) and the simplified DSP, makes the O-band an attractive solution for high throughput, low-complexity coherent transmission systems.

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